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Assignment 2: Process & Design
Product: BugFlow

Part II: Identifying Your Product & Target Audience

A. Define Your Product

1. What problem does your product website solve or address?

BugFlow addresses the problem of unclear and incomplete software bug reports. Regular users can usually tell when something went wrong with a website, application, game, or other software, but they may not know what information a developer needs in order to understand the problem. A report such as “the app does not work” tells the developer that there is an issue, but it does not explain what the user was doing, what steps led to the problem, what the user expected to happen, or what actually happened. BugFlow makes that communication easier by guiding the user through the important details in a simple way.

2. What is your product idea? How does your website support this idea and help solve the problem?

My product is called BugFlow. BugFlow is a web-based application that helps regular users create detailed and organized bug reports without needing a technical background. Instead of opening a traditional form and expecting the user to know terms such as “steps to reproduce,” “expected behavior,” or “system environment,” BugFlow asks normal questions one step at a time. The user can explain what application or website they were using, what device they were on, what they were trying to do, what happened, and whether they saw an error message. At the end, BugFlow organizes the answers into a structured report that can be copied, downloaded, or sent to a developer or support team.

3. What type of website are you building?

BugFlow will mainly be a web application. The site will also include informational pages that explain the product, show example reports, and describe its features, but the main purpose is interactive. The core experience is the Report a Bug flow, where users answer a series of questions and receive an organized report at the end.

4. What is your mission statement?

BugFlow’s mission is to make software bug reporting easier for everyone by helping regular users turn confusing software problems into clear, organized reports that developers can understand and use.

B. Profile of Target Audience

1. Who is the primary audience that your product is trying to reach?

The primary audience for BugFlow is people who use websites, applications, games, or other software and experience a problem but may not know how to properly report it. Secondary users

include software developers, QA testers, customer support teams, students, startups, and technology companies that need better information when software problems occur.

Demographics

- Age range: Approximately 18-45 years old. This range includes college students, working professionals, and frequent software users.
- Gender distribution: BugFlow is designed for users of all genders and does not target one gender more than another.
- Country: The first version will primarily target users in the United States, with the possibility of expanding internationally later.
- Urban or rural: Users may live in either setting, although many target users are likely to be in urban or suburban areas where technology-based jobs, colleges, and businesses are common.

Socioeconomic Details

- Average income: The audience can range from students with limited income to experienced technology professionals. A broad estimated range is about \$20,000 to \$100,000 or more per year.
- Education: Users may range from high school graduates and college students to people with bachelor's or graduate degrees.
- Marital or family status: No specific status is required. Users may be single, married, living with family, or raising children.
- Occupation: Possible users include college students, software developers, QA testers, customer support representatives, office workers, small business owners, and general software users.
- Hours worked per week: Working users will often work about 30-40 hours per week, while student schedules may vary.

Web Behavior

- Frequency: Most target users are expected to use the web every day.
- Devices: Users may access BugFlow from laptops, desktop computers, smartphones, or tablets. Because a software problem can happen on many devices, the site should work well across different screen sizes.

2. Fictional visitors from the target audience

Name	Sex	Age	Location	Occupation	Income	Web Use
Marcus Johnson	Male	21	Providence, RI	College Student	\$22,000	Uses the web several hours every day for school, entertainment, gaming, and social media.
Sarah	Female	29	Boston, MA	Customer	\$52,000	Uses

Name	Sex	Age	Location	Occupation	Income	Web Use
Martinez				Support Representative		websites and business applications throughout the workday.
Daniel Kim	Male	34	New York, NY	Software Developer	\$105,000	Uses the internet, development tools, and software platforms throughout the workday.

3. Reasons people would come to BugFlow

1. Report a software bug.
2. Create a more detailed and organized bug report.
3. Upload a screenshot of a software problem.
4. Learn what information developers need when a bug occurs.
5. Generate a completed bug report from simple answers.
6. Copy or download a completed report.
7. View examples of good bug reports.
8. Learn how BugFlow works.
9. Send organized bug information to a support team.
10. Use BugFlow as part of a QA or development workflow.

Tasks assigned to the fictional visitors

Marcus Johnson - College Student

Marcus notices that a school website freezes when he tries to upload an assignment. He opens BugFlow, selects Report a Bug, enters the website and device information, explains what he was trying to do, uploads a screenshot if available, generates the report, and then sends it to technical support.

Sarah Martinez - Customer Support Representative

Sarah receives a customer complaint saying that a mobile app “does not work.” She directs the customer to BugFlow, has the customer complete the guided questions, reviews the organized report, and forwards the useful technical details to the company’s development team.

Daniel Kim - Software Developer

Daniel receives a BugFlow report from a user. He reviews the device and operating system information, reads the user's steps, compares the expected result with the actual result, and then attempts to reproduce the bug so he can diagnose the cause.

Part III: Planning & Designing Your Product Website

A. Website Requirements

1. Purpose & Goals

Primary goal of the website

The primary goal of BugFlow is to help people create useful software bug reports without requiring them to understand software development or technical terminology. The site should make the process feel simple, guided, and manageable instead of overwhelming.

Most important action for users

The most important action is for a user to click Report a Bug and complete the guided reporting process. The homepage, navigation, and supporting content should all make it easy for a visitor to understand the product and begin a report.

2. Content & Features

Essential content and features

- Home: Explains what BugFlow is and gives users a clear Report a Bug call-to-action.
- Report a Bug: The main interactive feature. Users move through a series of simple questions.
- Example Reports: Shows the difference between weak and strong bug reports.
- Features: Explains guided reporting, screenshots, report generation, previews, copying, downloads, and missing-information warnings.
- About: Explains why BugFlow was created and the problem it is trying to solve.
- FAQ: Answers common questions such as what a bug report is, whether technical experience is required, what information should be included, and whether screenshots can be added.
- Generated Report: Displays the completed report in a structured format and gives the user options to copy, download, or start over.

Report a Bug process

Step	Main Question / Action
1 - Software	What application or website were you using?
2 - Device	What device and operating system were you using?
3 - Problem	What were you trying to do, and what actually happened?
4 - Reproduction	Walk us through what you did before the problem happened.
5 - Expected Result	What were you expecting to happen?

Step	Main Question / Action
6 - Evidence	Did you receive an error message? Can you upload a screenshot?
7 - Report	BugFlow organizes the information into a completed bug report.

Homepage highlights

The homepage will immediately explain what BugFlow does, show a short value statement, and display a large Report a Bug button. A three-step section will show visitors the basic flow: tell BugFlow what happened, let BugFlow organize the information, and then share the completed report. The homepage will also highlight that no technical knowledge is required.

3. Look & Feel

Design inspiration

Two websites that inspire the direction of BugFlow are Linear and GitHub. Linear is a useful reference because its interface is clean, modern, and organized without overwhelming the user. GitHub is a useful reference because issue reports organize software problems into structured information that developers can quickly scan. BugFlow will borrow the idea of clean structure while presenting questions in simpler language for nontechnical users.

Overall style

The overall style will be modern, clean, professional, simple, and technology-focused. I do not want the site to look overly technical because one of BugFlow's main goals is to make a technical process easier for normal users.

Color, font, and imagery preferences

The visual identity will use dark navy and blue as the main colors, with white and light gray backgrounds for readability. Red will be reserved for errors or warnings, while green can be used for successful completion states. The site will use a readable sans-serif interface font such as Inter, Arial, or Helvetica. Imagery will be limited and purposeful, using a friendly bug mascot, simple icons, and screenshots or interface examples when they help explain the product.

Color	Suggested Use	Hex
Dark Navy	Navigation, headings, footer	#0F172A
Blue	Primary buttons, links, progress indicators	#2563EB
White	Main background	#FFFFFF
Light Gray	Cards, form sections, secondary backgrounds	#F8FAFC
Red	Errors and warnings	#DC2626

Color	Suggested Use	Hex
Green	Successful report completion	#16A34A

B. Design & Prototyping

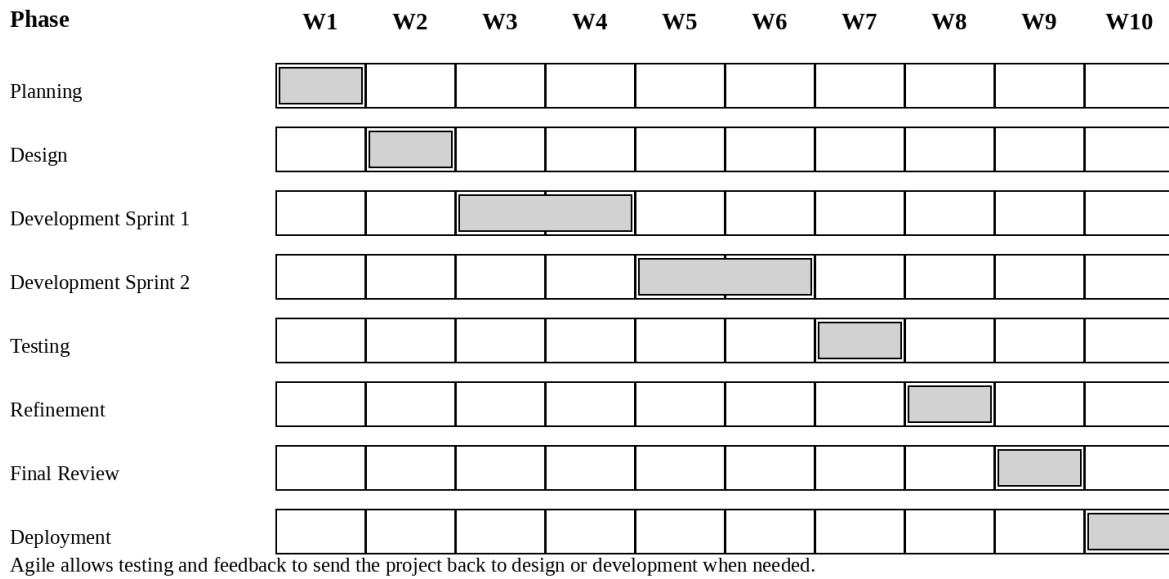
1. SDLC Approach & Timeline

I will use an Agile approach to build BugFlow. Agile fits this product because the website can be developed in smaller parts, tested, and improved as the semester continues. For example, I can build the basic navigation and homepage first, then create the guided bug-reporting form, test whether the questions make sense to users, and revise the wording or layout if something is confusing. This is useful for BugFlow because usability is one of the most important parts of the product.

Phase	Time	Main Tasks	Milestone
Planning	Week 1	Define the problem, audience, goals, and website requirements.	Product plan completed
Design	Week 2	Create logo, sitemap, color direction, and wireframes.	Website design completed
Development Sprint 1	Weeks 3-4	Build homepage, navigation, About, Features, and Example Reports pages.	Main site structure working
Development Sprint 2	Weeks 5-6	Build the interactive Report a Bug flow with HTML, CSS, and JavaScript.	Bug-reporting system working
Testing	Week 7	Test navigation, forms, responsive layout, and accessibility.	Major problems identified
Refinement	Week 8	Fix bugs and improve usability based on testing.	Final site improved
Final Review	Week 9	Run accessibility checks, review content, and complete final testing.	Website ready for submission
Deployment	Week 10	Upload final website and documentation to GitHub.	Project completed

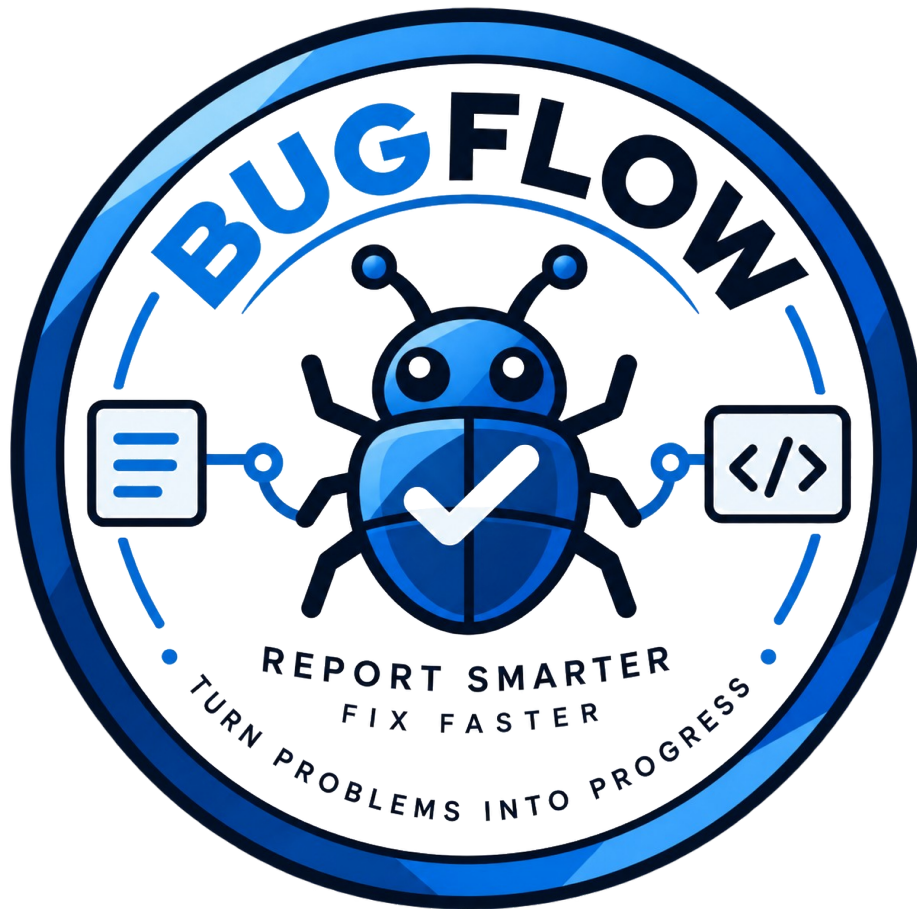
Visual timeline

BugFlow SDLC Timeline (Agile)



2. Original Logo

The BugFlow logo uses a bug mascot to represent software bugs, a checkmark to represent a completed and organized report, and connected document/code icons to represent the flow of information from a user problem to something a developer can use. The circular badge style was inspired by the provided reference, but the design uses BugFlow's own blue technology-focused identity and the phrases "Report Smarter," "Fix Faster," and "Turn Problems Into Progress."

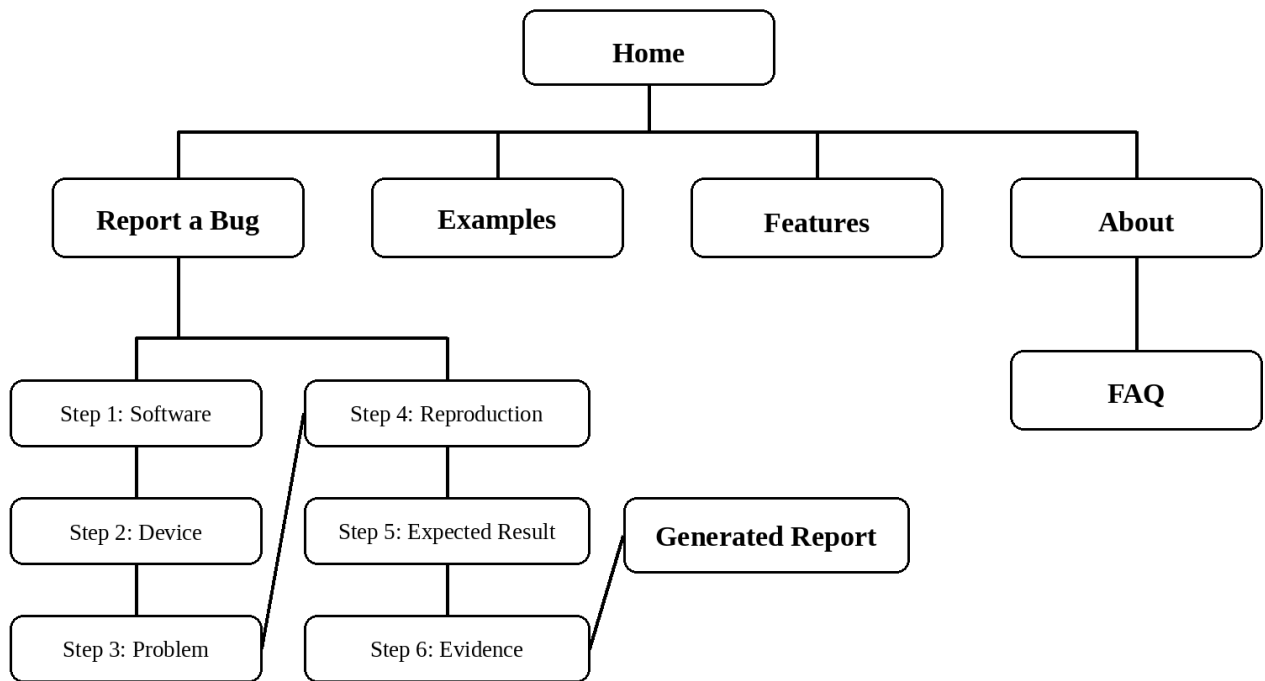


The logo is also exported separately as bugflow_logo.png with a transparent background so it can be used on the website.

3. Site Map

The site map below shows the main navigation and the internal flow of the Report a Bug feature. The top-level pages are Home, Report a Bug, Examples, Features, About, and FAQ. The reporting flow moves through the software, device, problem, reproduction, expected result, and evidence steps before generating the final report.

BugFlow Site Map



Primary navigation: Home | Report a Bug | Examples | Features | About | FAQ

4. Wireframes

The following wireframes show the planned layout for the homepage and two core pages. They focus on placement of the logo, navigation, content, forms, progress indicators, and main actions.

Homepage Wireframe

Homepage Wireframe

BUGFLOW LOGO[Home](#)[Report a Bug](#)[Examples](#)[Features](#)[About](#)[FAQ](#)

Turn software problems into reports developers can understand.

Answer a few simple questions and BugFlow organizes the details for you.

REPORT A BUG

How BugFlow Works

1. Tell Us What Happened

Answer simple questions about the problem.

2. BugFlow Organizes It

Your answers are placed into a structured report.

3. Share Your Report

Copy or download it for a developer or support team.

Why BugFlow?

- Easy to use
- No technical knowledge required
- Organized reports
- Mobile-friendly and accessible

Footer: [About](#) | [Contact](#) | [FAQ](#) | [GitHub](#)

Report a Bug Wireframe

Report a Bug Wireframe

BUGFLOW LOGO

[Home](#) [Report a Bug](#) [Examples](#) [Features](#) [About](#) [FAQ](#)

Report a Bug

Step 3 of 7

What were you trying to do?

Example: I was trying to upload my assignment...

What actually happened?

Describe the problem in your own words...

Did you receive an error message?

☐ Yes ☐ No

BACK

NEXT

Accessibility notes:

Keyboard navigation • clear labels • visible focus • error text + icons

Generated Report Wireframe

Generated Report Wireframe

BUGFLOW LOGO

[Home](#) [Report a Bug](#) [Examples](#) [Features](#) [About](#) [FAQ](#)

Your Bug Report Is Ready

Application / Website:

URI Brightspace

Device:

MacBook Air

Operating System:

macOS

What I was trying to do:

Upload an assignment file.

Steps before the problem:

1. Opened course page
2. Selected assignment
3. Chose upload
4. Selected file

Expected result:

The file should upload and appear in the submission area.

Actual result:

The page froze after selecting the file.

Error message:

No error message appeared.

Screenshot:

[Image preview]

COPY REPORT

DOWNLOAD

START OVER

5. Accessibility

Accessibility will be considered from the beginning of development instead of being added only after the site is finished. BugFlow is meant to simplify a technical process, so the site should also be easy to understand, navigate, and operate for users with different abilities.

Visual impairments

- Use strong color contrast between text, controls, and backgrounds.
- Provide alternative text for meaningful images and icons.
- Use proper semantic HTML headings so screen readers can understand page structure.
- Use readable font sizes and sufficient spacing.
- Do not use color as the only way to communicate an error, warning, or successful action. Error states will also include text and/or icons.

Hearing impairments

- BugFlow will not depend on audio to complete its main tasks.
- If video content is added later, it will include captions and a text transcript.
- Important instructions will always be available as visible text.

Motor impairments

- Make all major functions usable with a keyboard instead of requiring a mouse.
- Provide a visible focus indicator for links, buttons, and form fields.
- Use buttons and clickable areas that are large enough to select comfortably.
- Keep form controls in a logical tab order and avoid interactions that require precise mouse movement.

Cognitive impairments

- Use short, direct questions and avoid unnecessary technical terms.
- Break the bug-report form into smaller steps rather than showing one very large form.
- Use consistent navigation and button placement throughout the site.
- Show examples when a question may be confusing.
- Display a progress indicator so users know where they are in the process and how much is left.

Accessibility audit tools

- Google Lighthouse for automated accessibility checks and general web quality audits.
- WAVE Accessibility Evaluation Tool for contrast, labels, structure, and other common accessibility issues.
- Keyboard-only testing to confirm that all major actions can be completed without a mouse.
- Browser screen-reader testing to verify that form labels, headings, buttons, and status messages are announced correctly.
- HTML validation to help identify markup problems that may interfere with accessibility.